

3. Requirements Gathering

3.1 Stakeholder Analysis

Users

Role: Primary users of the system who search for restaurants and explore available options.

Needs:

- Search for restaurants easily
- View ratings and reviews
- Filter results based on preferences
- Access restaurant locations through maps
- Save favorite restaurants

Interest in System: High

Admin

Role: Responsible for managing and monitoring the system.

Needs:

- Manage restaurant data
- Update or remove incorrect information
- Moderate reviews and user-generated content
- Monitor system performance

Interest in System: High

Developers

Role: Responsible for building, maintaining, and improving the system.

Needs:

- Reliable API integration
- Scalable architecture
- Maintainable code structure
- Error handling and security implementation
- Performance optimization

Interest in System: High

3.2 User Stories

General Users

- As a user, I want to search for restaurants by name or location, so that I can find nearby places to eat.
- As a user, I want to filter restaurants by category, rating, and price range, so that I can choose the best option based on my preferences.
- As a user, I want to view restaurant details, so that I can see opening hours, menu information, and contact details.
- As a user, I want to read customer reviews and ratings, so that I can make informed decisions before visiting a restaurant.
- As a user, I want to view restaurants on a map, so that I can locate them easily.
- As a user, I want to save restaurants to a favorites list, so that I can access them later.
- As a user, I want to sort search results by rating or distance, so that I can find the most suitable restaurant faster.
- As a user, I want to create an account and log in, so that I can securely access and manage my saved preferences and favorite restaurants.
- As a user, I want to create an account, so that I can save my preferences and favorite restaurants.
- As a user, I want to log in to my account, so that I can access my saved data.

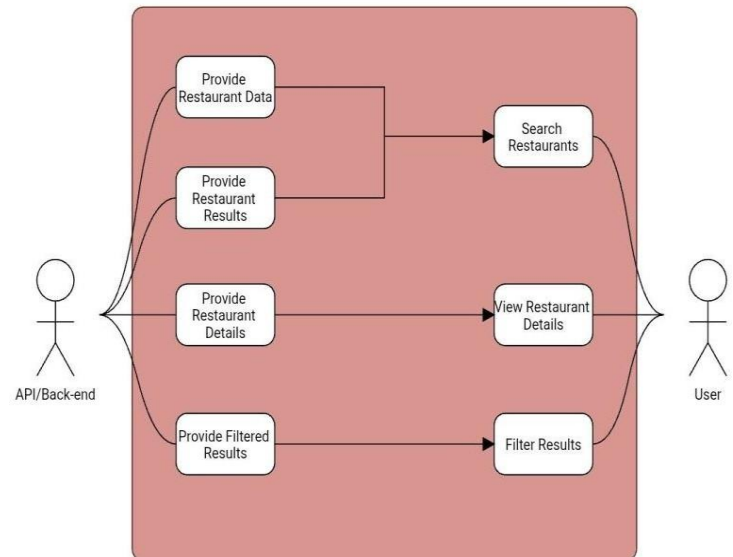
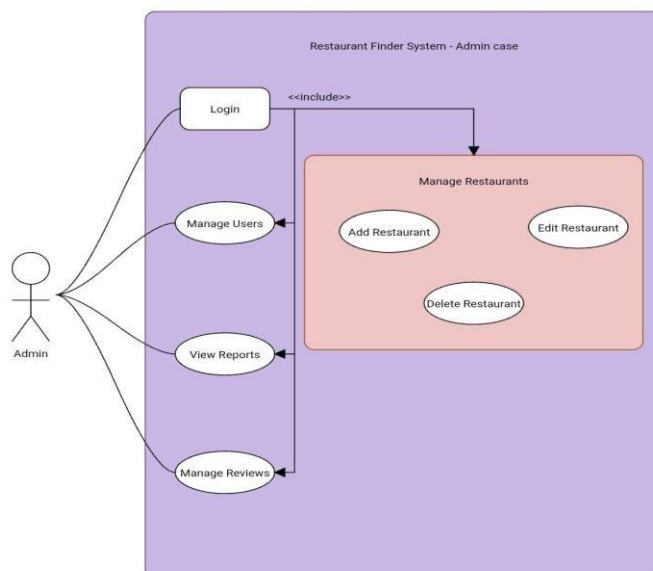
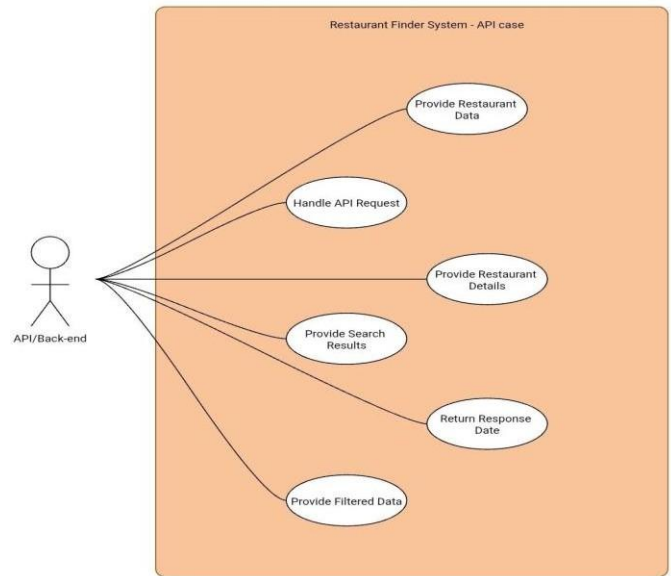
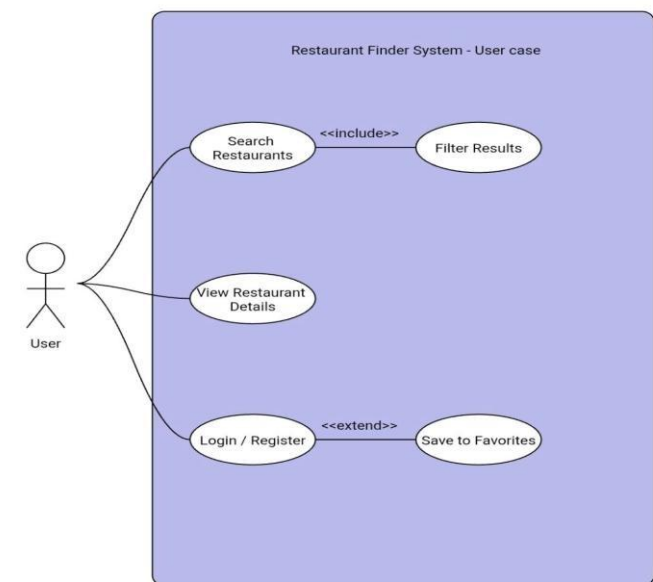
Developers

- As a developer, I want to integrate the Yelp API, so that restaurant data can be fetched dynamically.
- As a developer, I want to implement reusable React components, so that the system is maintainable and scalable.
- As a developer, I want to optimize search and filtering performance, so that users receive fast results.
- As a developer, I want to handle API errors properly, so that the system remains reliable when external services fail.
- As a developer, I want to maintain secure data handling, so that user information is protected.

Admin

- As an admin, I want to add restaurant records, so that the database stays updated.
- As an admin, I want to edit restaurant information, so that incorrect or outdated data can be corrected.
- As an admin, I want to remove invalid or duplicate restaurant entries, so that the system maintains data quality.
- As an admin, I want to manage user reviews, so that inappropriate content can be moderated.
- As an admin, I want to monitor system performance and usage, so that service quality can be maintained.

3.3 Use Cases



3.4 Functional Requirements

ID	Feature	Description
	Search Functionality	FR-01
The system shall allow users to search for restaurants by name, cuisine type, or location.		
	Restaurant Profiles	FR-02
The system shall display detailed information including menus, ratings, operating hours, and photos.		
	Advanced Filtering	FR-03
Users shall be able to filter search results by category, price range, and minimum rating.		

3.5 Non-Functional Requirements

Specification	Requirement	Category
	Response Time	Performance
The system should load pages and search results within less than 3 seconds.		
	Responsive Design	Usability
The user interface must be fully compatible with mobile, tablet, and desktop screens.		

User credentials and personal data must be encrypted using industry-standard protocols (e.g., SSL/HTTPS).	Data Protection	Security
The system should be operational 99.9% of the time to ensure constant access for users.	High Availability	Reliability